NAME: ____

- Each question is worth 5 points, 20 questions total and 1 bonus
- You should have 13 pages total
- You have the whole class for your attempt
- You should not have anything on you aside from your pen/ pencil and TI-30Xa calculator
- Do not use a red pen
- If you run out of space, you may continue your work on the blank reverse side or the last blank page as long as your work is clearly indicated
- Cancel working you do not want considered in your final answer. Show all working relevant to your final answer. Simplify and box all final answers
- If a numerical answer is not exact, express it using fractions/ radicals/ etc.

Q.1. Simplify the following expression:

$$\left(\frac{a^2 \cdot a^5}{b^{-2}}\right)^2 \cdot b^4$$

Q.2. Simplify the following expression (a, b may be any real numbers):

$$\sqrt{a^3 \cdot b^2} \cdot \sqrt{a^7 \cdot b^4}$$

Q.3. Simplify and rationalize the denominator:

$$\frac{x^2 \cdot y}{\sqrt[3]{y^5}}$$

Q.4. Rationalize the numerator:

$$\frac{5+\sqrt{x}}{5-\sqrt{x}}$$

Q.5. Simplify the following difference of rational expressions and **fully factor** the numerator and denominator in your final answer:

$$\frac{3}{x-2} - \frac{6}{x^2+x-6}$$

Q.6. Simplify the following quotient of rational expressions and **fully factor** the numerator and denominator in your final answer:

$$\frac{-2x^2+14x-20}{x-2} \div \frac{x^2-25}{3x}$$

Q.7. Find all real solutions of the following equation:

$$(2x - 2)^2 = (4x + 2) \cdot (x - 4)$$

Q.8. 550 people attended the premiere of a film. Each adult ticket cost \$10 and each child ticket cost \$5. If box office revenue totalled $\$5,000$, how many adults and how many children attended the premiere? (**Hint:** you will need two variables and two equations)

Q.9. Simplify the following quotient of complex numbers into a single complex number (of the form $a + bi$ with a, b real):

$$\frac{4 - 3i}{1 - 2i}$$

Q.10. State the quadratic formula for the general equation $ax^2 + bx + c = 0$. Then use this formula to find all **complex** solutions of the equation $-4x^2 + 8x - 5 = 0$:

Q.11. Alice starts at the beginning of a runner's path and runs at a constant rate of 5 mi/hr. 20 minutes later Bob starts running from the same starting point at a constant rate of 8 mi/hr.
How long **in minutes** will it take for Bob to catch up to Alice? (it helps to draw a diagram on the reverse side of the page!)

Q.12. Find all real solutions of the following equation using the zero-factor theorem:

$$2x^3 - 3x^2 - 8x + 12 = 0$$

Q.13. Dina is taking a class in which her total grade is determined only by her scores on 3 midterm tests and 1 final exam. The final exam is worth half her total grade and the midterm tests all have equal weight. On her midterm tests, Dina scored $94\%, 83\%$, and 87%, respectively. What percentage does Dina need to get on the final exam to achieve a total grade of exactly 90%?

Q.14. Find all **real** solutions of the equation:

$$x^4 - x^2 - 12 = 0$$

Q.15. Find all real solutions of the following equation. Make sure to **plug them back in to the equation** and check they are correct:

$$x - 1 = -\sqrt{7 - x}$$

Q.16. Find all real solutions of the equation:

$$\sqrt[3]{x^2 - 6x} = -2$$

Q.17. The equation for the total amount accrued via simple interest is $A = P + Prt$ where P denotes the principal. Use this equation to express P in terms of the other variables, A, r, t (total amount, rate of interest, and time, respectively).

Q.18. Find all real solutions to the inequality and write your answer in interval notation (draw a sketch of your solution set if confused):

$$-5 \leq -3x + 1 < 10$$

Q.19. State the **domain** of this inequality. Then find all real solutions to the inequality and write your answer in interval notation (draw a sketch of your solution set if confused):

$$\frac{4}{|x+1|} > 1$$

Q.20. Find all real solutions to the inequality and write your answer in interval notation (draw a sketch of your solution set if confused):

$$\frac{x(x-2)}{x+3} < 0$$

BONUS: We mentioned that for a quadratic equation, $ax^2 + bx + c = 0$, where a, b, c are real coefficients, there are only three possibilities for the roots of the equation: we may have two distinct real solutions, one unique real solution, or two distinct complex solutions.
But why can't we have just one unique non-real complex solution?
e.g. why isn't there a quadratic equation with real coefficients whose only root is i? Explain.

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